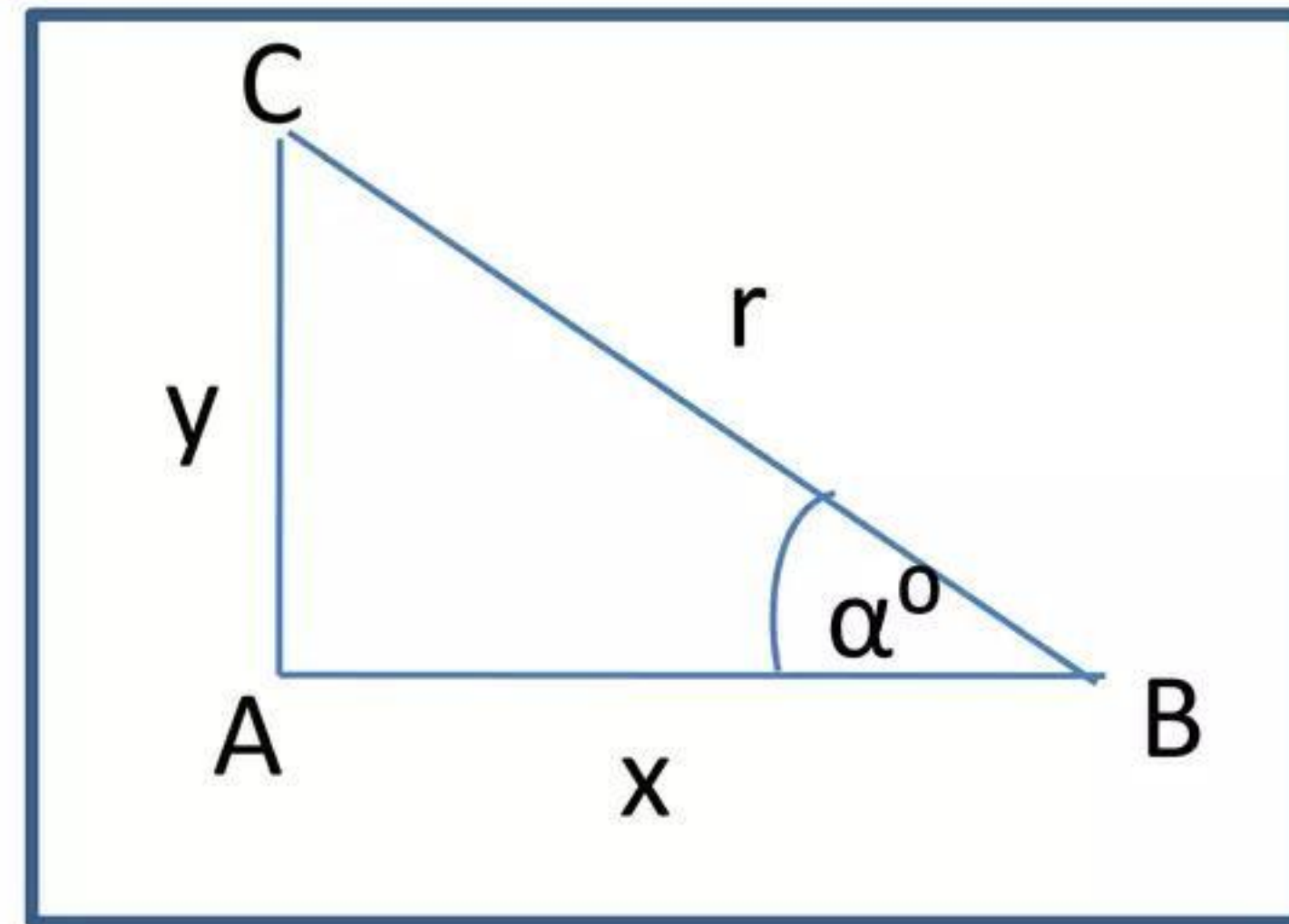


IDENTITAS TRIGONOMETRI DASAR



“Mengingat Kembali”



$$\sin \alpha^{\circ} = \frac{y}{r}$$

$$\cos \alpha^{\circ} = \frac{x}{r}$$

$$\tan \alpha^{\circ} = \frac{y}{x}$$



Kuadran 2

Tanda		
sin	cos	tan
+	-	-

Kuadran 1

Tanda		
sin	cos	tan
+	+	+

Kuadran 3

Tanda		
sin	cos	tan
-	-	+

Kuadran 4

Tanda		
sin	cos	tan
-	+	-

Identitas trigonometri dasar merupakan hubungan kebalikan

$$\sin \alpha^\circ = \frac{1}{\operatorname{cosec} \alpha^\circ} \quad \text{atau} \quad \operatorname{cosec} \alpha^\circ = \frac{1}{\sin \alpha^\circ} = \frac{1}{\frac{y}{r}} = \frac{r}{y}$$

$$\cos \alpha^\circ = \frac{1}{\sec \alpha^\circ} \quad \text{atau} \quad \sec \alpha^\circ = \frac{1}{\cos \alpha^\circ} = \frac{1}{\frac{x}{r}} = \frac{r}{x}$$

$$\tan \alpha^\circ = \frac{1}{\cot \alpha^\circ} \quad \text{atau} \quad \cot \alpha^\circ = \frac{1}{\tan \alpha^\circ} = \frac{1}{\frac{y}{x}} = \frac{x}{y}$$

Identitas trigonometri dasar merupakan hubungan perbandingan

$$\tan \alpha^{\circ} = \frac{\sin \alpha^{\circ}}{\cos \alpha^{\circ}}$$

$$\cot \alpha^{\circ} = \frac{1}{\tan \alpha^{\circ}} = \frac{1}{\frac{\sin \alpha^{\circ}}{\cos \alpha^{\circ}}} = \frac{\cos \alpha^{\circ}}{\sin \alpha^{\circ}}$$

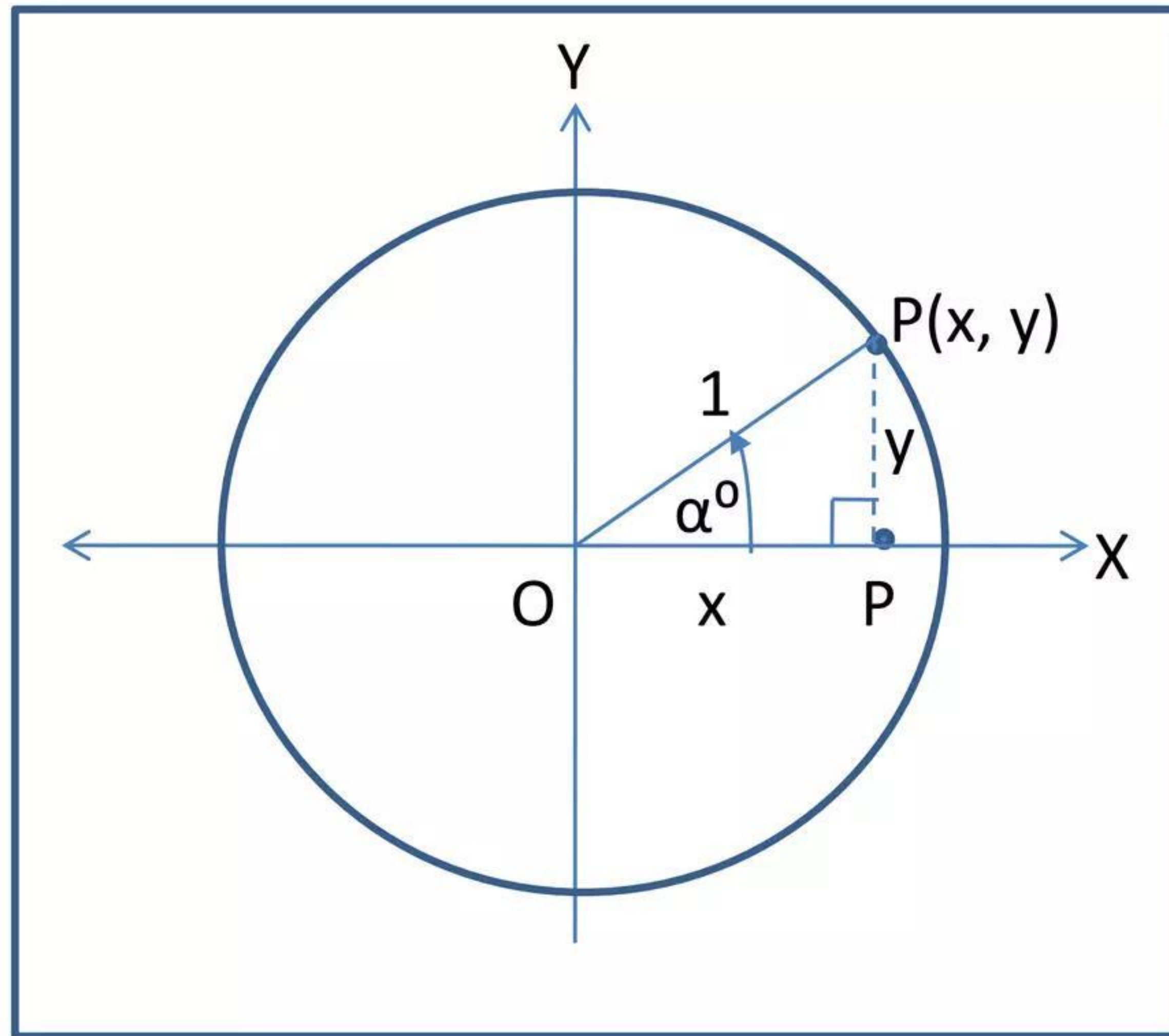
$$(OP')^2 + (PP')^2 = (OP)^2$$

$$x^2 + y^2 = 1$$

$$\cos \alpha^\circ = \frac{x}{1} \text{ dan } \sin \alpha^\circ = \frac{y}{1}$$

Karena $x = \cos \alpha^\circ$ dan $y = \sin \alpha^\circ$ maka diperoleh

$$\cos^2 \alpha^\circ + \sin^2 \alpha^\circ = 1$$



Jika kedua ruas persamaan $x^2 + y^2 = 1$ dibagi dengan x^2 , maka diperoleh:

$$\frac{x^2}{x^2} + \frac{y^2}{x^2} = \frac{1}{x^2} \quad \Leftrightarrow \quad 1 + \left(\frac{y}{x}\right)^2 = \left(\frac{1}{x}\right)^2$$

Substitusi $\frac{y}{x} = \tan \alpha^\circ$ dan $\frac{1}{x} = \sec \alpha^\circ$ ke persamaan di atas ,
maka diperoleh

$$1 + \tan^2 \alpha^\circ = \sec^2 \alpha^\circ$$

Sekarang jika kedua ruas persamaan $x^2 + y^2 = 1$ dibagi dengan y^2 , maka diperoleh

$$\frac{x^2}{y^2} + \frac{y^2}{y^2} = \frac{1}{y^2} \quad \Leftrightarrow \quad 1 + \left(\frac{x}{y}\right)^2 = \left(\frac{1}{y}\right)^2$$

Substitusi $\frac{x}{y} = \cot \alpha^\circ$ dan $\frac{1}{y} = \operatorname{cosec} \alpha^\circ$ ke persamaan di atas ,
maka diperoleh

$$1 + \cot^2 \alpha^\circ = \operatorname{cosec}^2 \alpha^\circ$$

Identitas trigonometri dasar yang diperoleh dari teorema Pythagoras

$$\sin^2 \alpha^\circ + \cos^2 \alpha^\circ = 1$$

$$1 + \tan^2 \alpha^\circ = \sec^2 \alpha^\circ$$

$$1 + \cot^2 \alpha^\circ = \operatorname{cosec}^2 \alpha^\circ$$

Contoh soal

Diketahui $\sin \alpha^\circ = \frac{3}{5}$ dan $0 < \alpha < 90$. Hitunglah:

a) $\cos \alpha$

b) $\tan \alpha$

Jawab:

a) Dengan menggunakan rumus:

$$\sin^2 \alpha^\circ + \cos^2 \alpha^\circ = 1$$

$$\Leftrightarrow \cos^2 \alpha^\circ = 1 - \sin^2 \alpha^\circ$$

$$\Leftrightarrow \cos^2 \alpha^\circ = 1 - \left(\frac{3}{5}\right)^2$$

$$\Leftrightarrow \cos^2 \alpha^\circ = \frac{16}{25}$$

$$\Leftrightarrow \cos \alpha^\circ = -\frac{4}{5} \text{ atau } \cos \alpha^\circ = \frac{4}{5}$$

Karena $0 < \alpha < 90$ (terletak di kuadran I), maka diambil $\cos \alpha = \frac{4}{5}$

b) Dengan menggunakan rumus perbandingan:

$$\tan \alpha^\circ = \frac{\sin \alpha^\circ}{\cos \alpha^\circ}$$

$$\Leftrightarrow \tan \alpha^\circ = \frac{\frac{3}{5}}{\frac{4}{5}} = \frac{3}{4}$$

2) Buktikan bahwa $\sin^2 \alpha^\circ + \sin^2 \alpha^\circ \cos^2 \alpha^\circ + \cos^4 \alpha^\circ = 1$

Jawab:

Kita ubah bentuk ruas kiri:

$$\begin{aligned}\sin^2 \alpha^\circ + \sin^2 \alpha^\circ \cos^2 \alpha^\circ + \cos^4 \alpha^\circ &= \sin^2 \alpha^\circ (1 + \cos^2 \alpha^\circ) + \cos^4 \alpha^\circ \\ &= (1 - \cos^2 \alpha^\circ)(1 + \cos^2 \alpha^\circ) + \cos^4 \alpha^\circ \\ &= 1 - \cos^4 \alpha^\circ + \cos^4 \alpha^\circ \\ &= 1\end{aligned}$$

Ruas kiri = Ruas kanan

Jadi, terbukti bahwa $\sin^2 \alpha^\circ + \sin^2 \alpha^\circ \cos^2 \alpha^\circ + \cos^4 \alpha^\circ = 1$

THANK
YOU

